

P1 : Drought Severity Mapping using Vegetation Condition Index(VCI)

**Project Report Submitted in fulfillment of the
Requirements for the Award of the Internship of Summer
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Geospatial Intelligence

By

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Project 1 :Drought Severity Mapping using Vegetation Condition Index(VCI) – Jaisalmer Region

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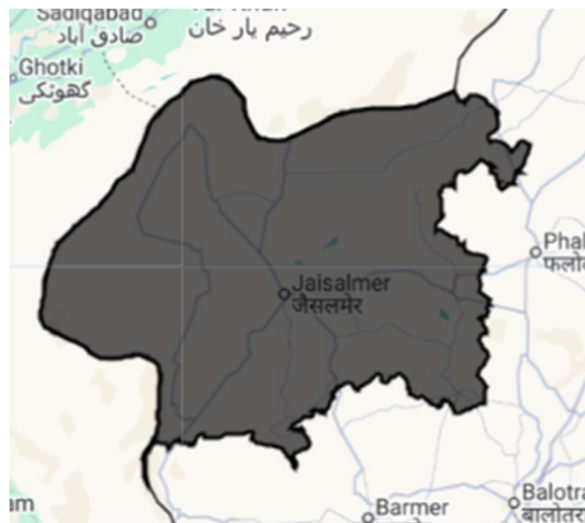
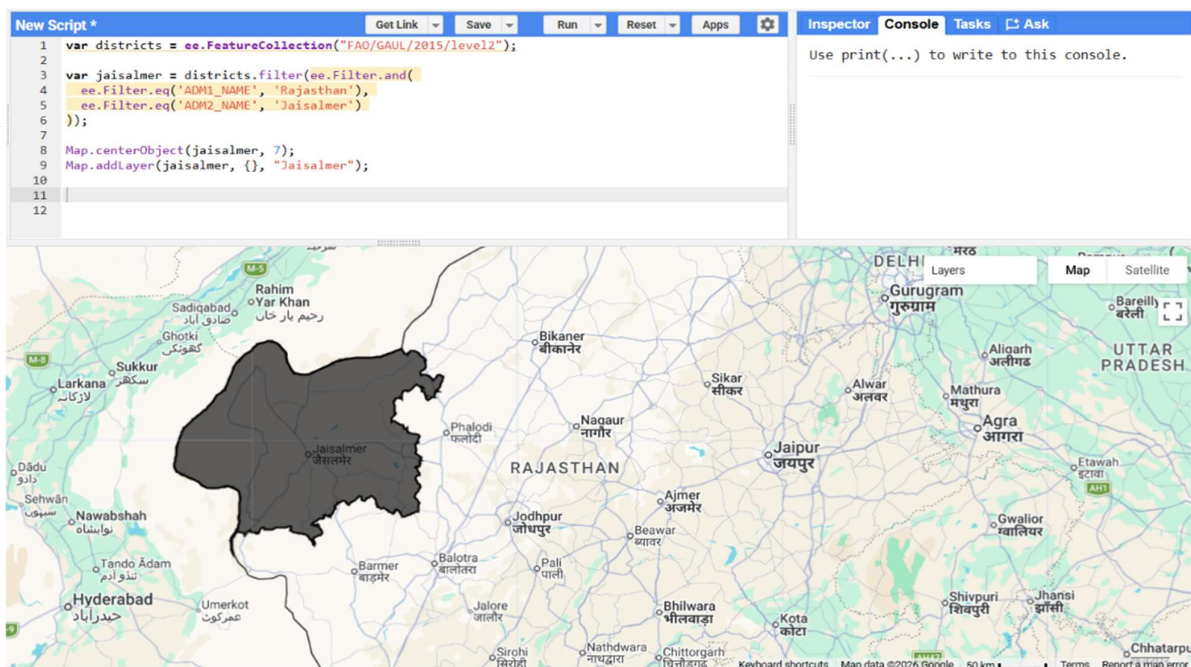
Date of Submission : 1 July,2026

Objective

The objective of this project is to analyze vegetation health in Jaisalmer using the Vegetation Condition Index (VCI). This helps in understanding drought conditions and vegetation stress over time using satellite data.

Study Area

The study area selected for this project is Jaisalmer, located in the state of Rajasthan, India. It is a desert region known for its dry climate, low rainfall, and sparse vegetation. Due to these conditions, it is suitable for studying vegetation stress and drought patterns.



Data Used

Satellite Data:

- MODIS NDVI (dataset)
 - Sentinel-2

Required Bands:

- Near Infrared (NIR)
 - Red Band

Software Used:

- Google Earth Engine

Methodology

The following steps were performed in Google Earth Engine:

- Defined the Area of Interest (Jaisalmer)
- Loaded Sentinel-2 dataset
- Filtered data based on date and cloud cover
- Calculated NDVI using Red and NIR bands

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

- Higher NDVI values indicate healthier vegetation, lower values indicate stressed vegetation

- Calculated minimum and maximum NDVI values
- Applied the VCI formula:

$$\text{VCI} = (\text{NDVI} - \text{NDVI}_{\min}) / (\text{NDVI}_{\max} - \text{NDVI}_{\min}) \times 100$$

- Classification

VCI < 20 : Extreme Drought

20 – 40 : Moderate Drought

40 – 60 : Mild Drought

> 60 : Normal Vegetation

- Visualized the result using a color palette
- Exported the output image

New Script *

Get Link

Save

Run

Reset

Apps

Inspector

Console

Tasks

Ask

```

21 var ndviMin = ndvi.reduce(ee.Reducer.min());
22 var ndviMax = ndvi.reduce(ee.Reducer.max());
23
24
25 var currentNDVI = ndvi
26   .filterDate('2022-01-01', '2022-12-31')
27   .mean()
28   .rename('NDVI');
29
30 print("Current NDVI:", currentNDVI);
31
32 // VCI calculation
33 var vci = currentNDVI.subtract(ndviMin)
34   .divide(ndviMax.subtract(ndviMin))
35   .multiply(100)
36   .clip(jaisalmer);
37
38 print(ndvi.size());
39
40 // Display
41 var palette = ['red', 'orange', 'yellow', 'green', 'darkgreen'];
42
43 Map.addLayer(vci, {min: 0, max: 100, palette: palette}, 'VCI Jaisalmer');
44
45 var stats = vci.reduceRegion({
46   reducer: ee.Reducer.mean(),
47   geometry: jaisalmer,
48   scale: 250,
49   maxPixels: 1e13
50 });
51
52 print(stats);
53
54 // Create legend panel
55 var legend = ui.Panel({
56   style: {
57     position: 'bottom-left',
58     padding: '8px 15px'
59   }
60 });
61

```

Use print(...) to write to this console.

Current NDVI:

JSOI

Image (1 band)

JSOI

type: Image

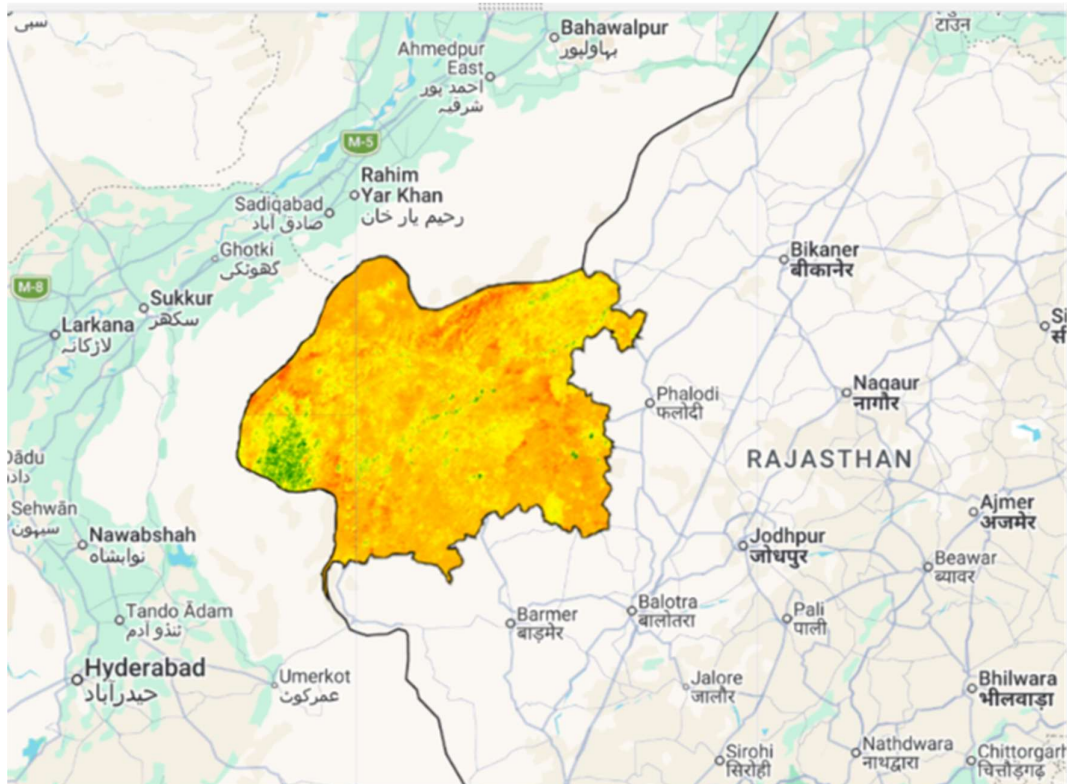
bands: List (1 element)

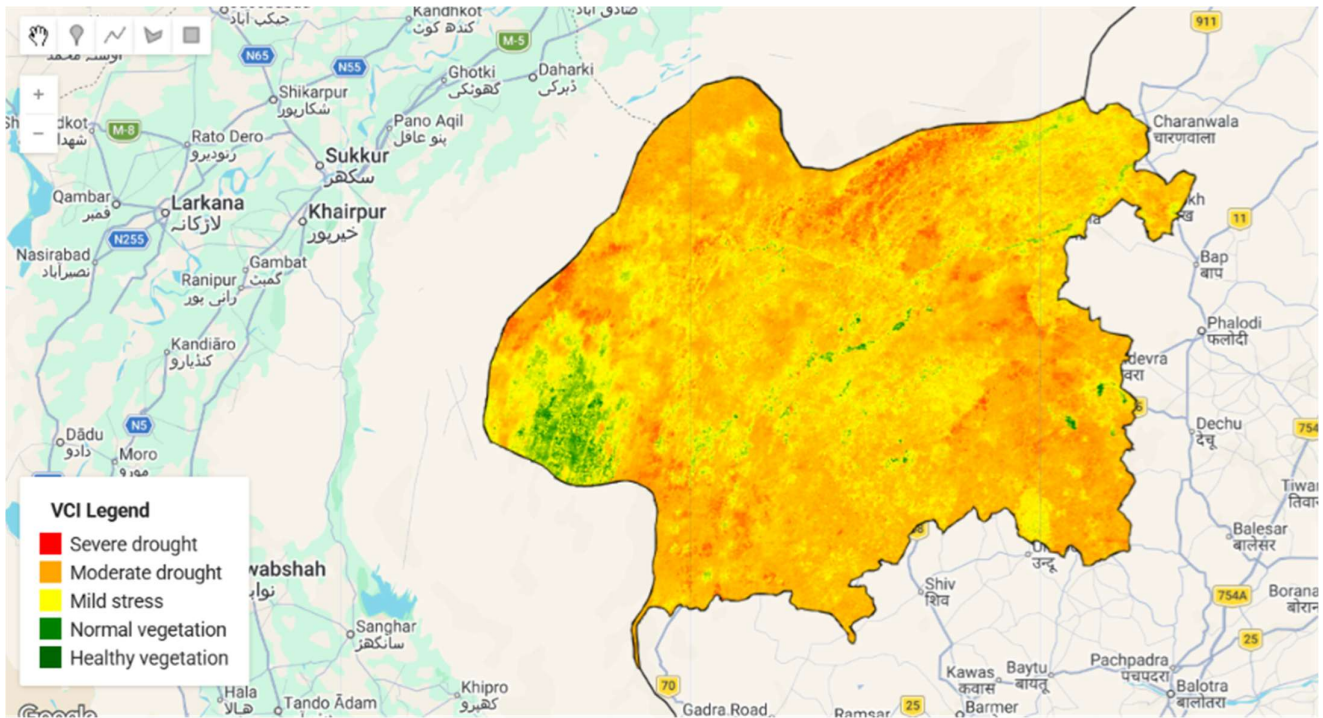
187

Object (1 property)

JSOI

NDVI: 34.29728215025003





A color palette was used to represent vegetation conditions:

Red → Very low vegetation (high stress)

Orange/Yellow → Moderate vegetation

Green → Healthy vegetation

This helps in easily identifying drought-prone areas.

Results

The VCI output shows that most parts of Jaisalmer have low to moderate vegetation conditions. Large areas appear in red and yellow shades, indicating vegetation stress. Some small patches of green areas are visible, which may represent irrigated land or seasonal vegetation.

Conclusion

This project demonstrates how satellite data can be used to monitor vegetation conditions. The results show that Jaisalmer has mostly low vegetation due to its desert climate. VCI is useful for identifying drought-prone areas and can help in environmental monitoring.

References

- Google Earth Engine
- MODIS NDVI Dataset (GEE)
- Google Earth Engine Documentation
- India Space Academy Summer Internship Project Guidelines
- Sentinel-2 Dataset
- NDVI/VCI formulas (class notes)